Safexpress Centre for Data, Learning and Decision Sciences

**Experimental AI Subseasonal-to-Seasonal (S2S) Forecast Guidance for India**

**Forecast Initialization:** 2 September 2026
**Forecast Period:** 2–29 September 2026
**Forecast Range:** Weeks 1–4
**Reference Dataset:** ERA5 reanalysis/model climatology, 2002–2021
**Forecast Product:** IFS operational initialization, 100-member ensemble mean
**Status:** Experimental Research Guidance

The 2 September product covers four forecast weeks: 2–8 September, 9–15 September, 16–22 September, and 23–29 September 2026. The rainfall, temperature, and 850-hPa circulation anomalies are calculated relative to the corresponding 2002–2021 model climatology.

## Executive Summary

The 2 September 2026 S2S initialization indicates a relatively active and northward-displaced monsoon rainfall configuration during Week 1, followed by a marked weakening of the continental wet signal during Weeks 2–3. By Week 4, the anomaly pattern becomes weaker and more spatially heterogeneous, with some indication of renewed above-normal rainfall over southern peninsular India and the adjoining Indian Ocean. During Week 1, 2–8 September, the most prominent rainfall signal is a broad positive anomaly extending across north-central India and into the eastern Indo-Gangetic/eastern Indian sector. Positive rainfall anomalies extend from portions of eastern Rajasthan and Madhya Pradesh toward Uttar Pradesh, Bihar, Jharkhand, West Bengal and adjoining areas, with additional enhancement along parts of the Himalayan foothill region. In contrast, large parts of the western and southern peninsula are forecast to receive below-normal rainfall. The 850-hPa circulation anomaly shows enhanced low-level flow across the Arabian Sea and central India, dynamically supporting the north-central wet anomaly on week 1. The corresponding temperature signal is predominantly cooler than normal across northern, central and eastern India, whereas southern peninsular India is warmer than normal. A major transition occurs during Week 2, 9–15 September. Positive rainfall anomalies over continental India largely dissipates, leaving most of western, central and peninsular India near to below normal. Limited positive anomalies remain over portions of Northeast India, while a much stronger wet anomaly develops farther south over the equatorial and southern Indian Ocean. At 850 hPa, anomalous easterly flow becomes increasingly evident across the Arabian Sea and peninsular sector, accompanied by widespread negative wind-speed anomalies mainly due to anomalous anticyclone over north-west India. This suggests a substantial weakening of the climatological southwest-monsoon flow into India. During Week 3, 16–22 September, below-normal rainfall becomes widespread across much of northern, central and eastern India. Positive anomalies are largely displaced toward Northeast India, portions of the southern peninsula, and the southern Indian Ocean. The low-level circulation exhibits its clearest withdrawal-like configuration during this period: anomalous

easterly flow and weaker-than-normal winds dominate much of the peninsula and adjoining seas. Temperatures simultaneously become warmer than normal across most of central, western and peninsular India. By Week 4, 23–29 September, the signal becomes less consistent with a simple coherent progression toward withdrawal. Rainfall anomalies are weak or negative over northern India and the Himalayan foothill belt, but positive anomalies become more prominent over southern peninsular India, the southern Bay of Bengal and adjoining Indian Ocean, while some positive rainfall signal persists near Northeast India. The 850-hPa anomaly field also shows stronger anomalous low-level flow south of India. Consequently, Week 4 is better characterized as a weaker and reorganized late-monsoon circulation rather than simply a continuation of the strong Week 3 weakening. Overall, the updated forecast suggests that the most distinct withdrawal-like transition occurs during Weeks 2–3, following the active north-central rainfall pattern in Week 1. The forecast does not establish the formal timing of southwest-monsoon withdrawal, particularly because the Week 4 forecast contains indications of renewed rainfall and circulation activity over southern India and adjacent oceanic regions.

## 1. Methodology

The forecast is based on the 2 September 2026 IFS operational initialization and a 100-member ensemble mean. Weekly fields are evaluated for rainfall, 2-m temperature and 850-hPa winds over four consecutive seven-day forecast periods. Rainfall and temperature anomalies represent departures from the model's typical conditions for the corresponding season and forecast lead during 2002–2021. The rainfall-anomaly product therefore indicates whether a location is expected to be wetter or drier than normal rather than whether rainfall will occur at all. The 850-hPa anomaly product represents both the anomaly in weekly mean wind speed and anomalous U/V wind components relative to 2002–2021.

## 2. Rainfall Forecast

**2.1 Week 1: 2–8 September 2026 -** Week 1 contains the strongest organized continental wet anomaly in the 2 September forecast cycle. A broad positive rainfall-anomaly zone extends across north-central India and eastward through the Indo-Gangetic/eastern Indian corridor. Particularly pronounced positive anomalies occur over portions of northern and eastern Madhya Pradesh and adjoining north-central India, with another wet center extending into eastern India. Positive anomalies also extend along portions of the Himalayan foothill region.

In contrast, below-normal rainfall dominates much of the southern peninsula. Negative anomalies extend across portions of Maharashtra, Karnataka, Tamil Nadu and adjoining areas, together with substantial parts of the central and southern Bay of Bengal.

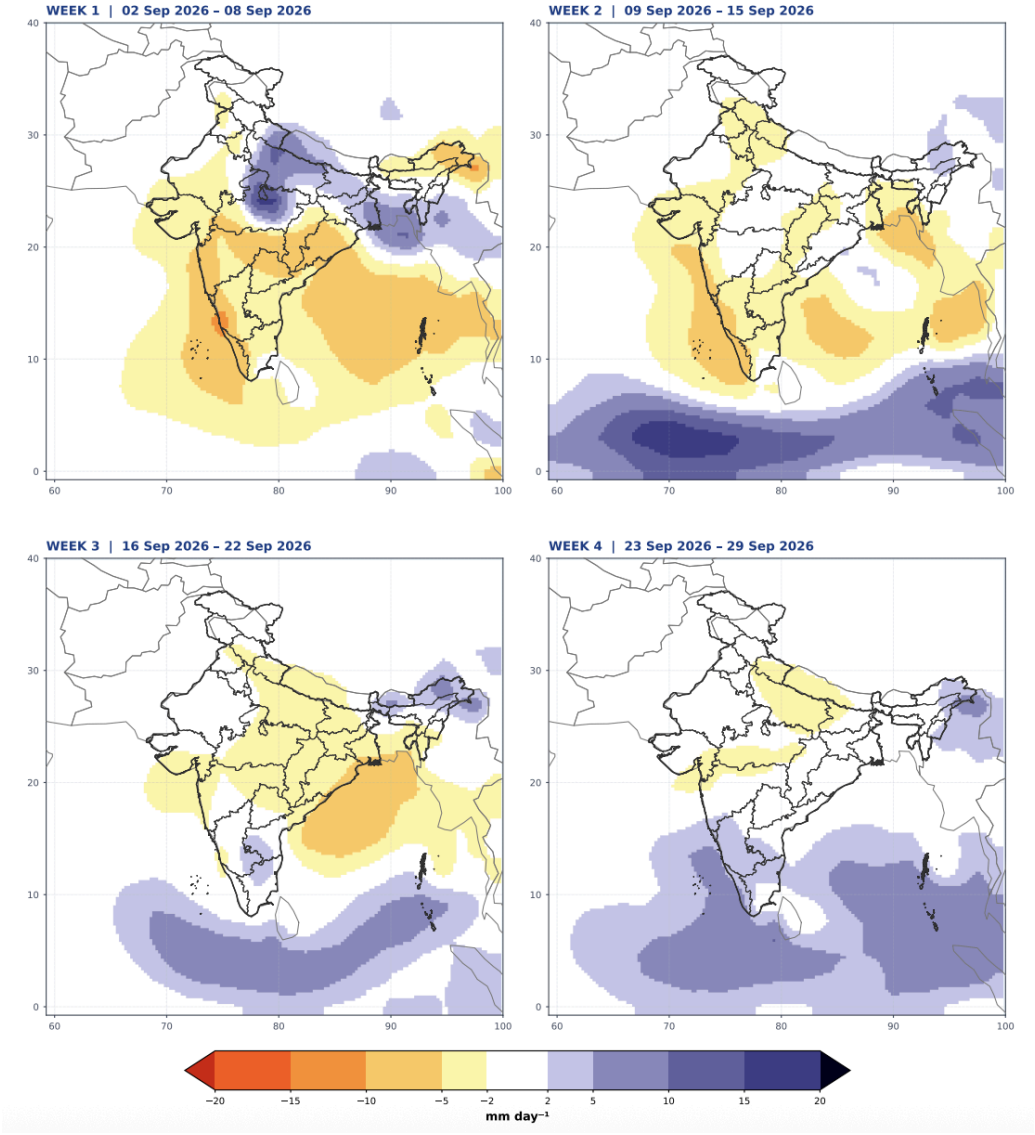



**Figure 1.** Weekly mean rainfall anomaly over India for Weeks 1–4, IFS operational initialization dated 2 September 2026. Panels represent (a) 2–8 September, (b) 9–15 September, (c) 16–22 September, and (d) 23–29 September 2026. Anomalies represent departures from the model's typical rainfall for the corresponding season and forecast lead during 2002–2021.

Absolute rainfall nevertheless remains appreciable over several parts of central and eastern India and is particularly high toward eastern and northeastern India. Thus, the Week 1 pattern represents a northward and eastward concentration of monsoon rainfall, rather than a uniformly active monsoon across the country. The rainfall-anomaly map for the 2 September initialization shows this strong north-central/eastern wet corridor clearly, while the peninsula is predominantly below normal.

**2.2 Week 2: 9–15 September 2026 -** A major weakening of the continental rainfall signal occurs during Week 2. Most of India becomes near normal to drier than normal, with negative rainfall anomalies evident over western India, parts of the west coast, the peninsula, eastern India and adjoining Bay of Bengal. The strong north-central wet anomaly present during Week 1 is no longer evident. Some positive anomalies remain over portions of Northeast India, but the more prominent positive rainfall anomaly is displaced southward into the equatorial and southern Indian Ocean. This represents a pronounced change from Week 1 and indicates a substantial reduction in organized monsoon rainfall over the Indian landmass.

**2.3 Week 3: 16–22 September 2026 -** Week 3 continues the broadly suppressed continental rainfall regime. Below-normal rainfall extends across much of western, central, northern and eastern India, with a coherent dry anomaly covering a substantial portion of the central and eastern peninsula. Positive anomalies are substantially more localized. They occur over parts of **Northeast India** and portions of southern peninsular India, particularly toward the southern interior, while a broad positive anomaly remains over the southern Indian Ocean. The spatial separation between the wet southern-oceanic anomaly and the predominantly dry central Indian landmass is an important feature of the Week 3 forecast.

**2.4 Week 4: 23–29 September 2026 -** By Week 4, the anomaly pattern becomes weaker and more heterogeneous. Negative rainfall anomalies remain evident across parts of northern India and the Himalayan foothill/Indo-Gangetic region. In contrast, much of central and eastern India is closer to climatology than during Week 3. More importantly, positive anomalies expand across southern peninsular India and the adjacent Arabian Sea–Indian Ocean–southern Bay of Bengal sector. Positive anomalies also remain near portions of Northeast India. Consequently, Week 4 should not be interpreted as simply a stronger version of the Week 3 dry pattern. Instead, the forecast suggests a possible **southward redistribution of rainfall activity**, although confidence at this lead time is substantially lower.

**3. Wind and Circulation Forecast**

The 850-hPa wind guidance provides important dynamical support for the rainfall evolution. The source product shows weekly ensemble-mean winds separately from weekly wind-speed and vector-component anomalies.

**Week 1 -** During Week 1, a substantial positive low-level wind anomaly extends from the Arabian Sea into western and central India. Anomalous flow is predominantly directed eastward across the central monsoon corridor, indicating stronger-than-normal westerly components and enhanced low-level transport toward central and eastern India. This circulation is dynamically consistent with the pronounced positive rainfall anomalies extending across north-central and eastern India. The circulation therefore remains distinctly monsoon-active during 2–8

September and does not support an interpretation of widespread monsoon withdrawal during Week 1.

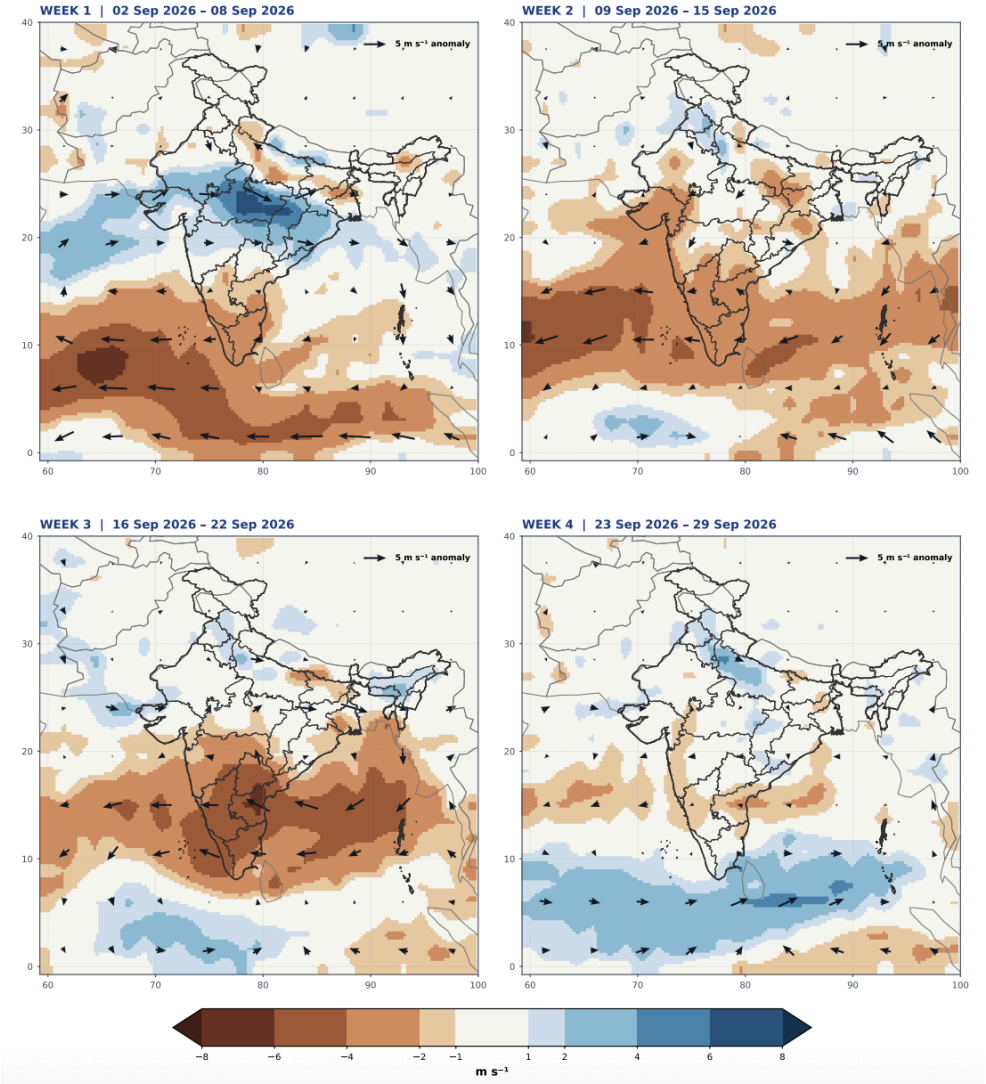



**Figure 2.** Weekly mean wind anomaly over India for Weeks 1–4, IFS operational initialization dated 2 September 2026. Panels represent (a) 2–8 September, (b) 9–15 September, (c) 16–22 September, and (d) 23–29 September 2026. Anomalies represent departures from the model's typical wind for the corresponding season and forecast lead during 2002–2021.

**Week 2 -** The circulation changes markedly during Week 2. The Week 1 positive wind-speed anomalies across central India weaken substantially, while negative wind-speed anomalies become widespread across the Arabian Sea, southern India and adjoining waters. Anomalous easterly components become prominent over the Arabian Sea and peninsula. Because

climatological southwest-monsoon flow is predominantly westerly/southwesterly, these easterly anomalies are indicative of a weakening of the normal monsoon circulation. The simultaneous disappearance of the Week 1 continental wet anomaly provides further support for this interpretation.

**Week 3 -** The withdrawal-like circulation signal becomes strongest during Week 3. A broad region of substantially weaker-than-normal low-level winds covers southern and peninsular India and extends into the Bay of Bengal and Arabian Sea. Anomalous easterly flow is widespread across the peninsula and adjacent seas. This configuration is considerably less favorable for sustained southwest-monsoon moisture transport into central India and is dynamically consistent with the widespread negative rainfall anomalies over western, central and eastern India.

**Week 4 -** Week 4 shows a partial reorganization of the circulation. Low-level anomalies over much of India itself become weaker than during Week 3. At the same time, a band of positive wind-speed anomalies and anomalous westerly flow appears south of the Indian peninsula. This southern circulation feature coincides with the emergence of positive rainfall anomalies over southern India and adjoining oceanic regions. Therefore, the Week 4 circulation should be interpreted cautiously. The forecast still depicts a weak late-season monsoon circulation over much of continental India, but the signal is not a simple continuation of the Week 3 weakening. The 850-hPa wind-anomaly panels for all four weeks are provided on page 6 of the forecast product.

**4. Temperature Forecast**

The temperature-anomaly guidance shows a clear transition from relatively cool north-central/eastern conditions during Week 1 toward widespread warming during Weeks 2–4. Temperature anomalies are relative to the same 2002–2021 reference period.

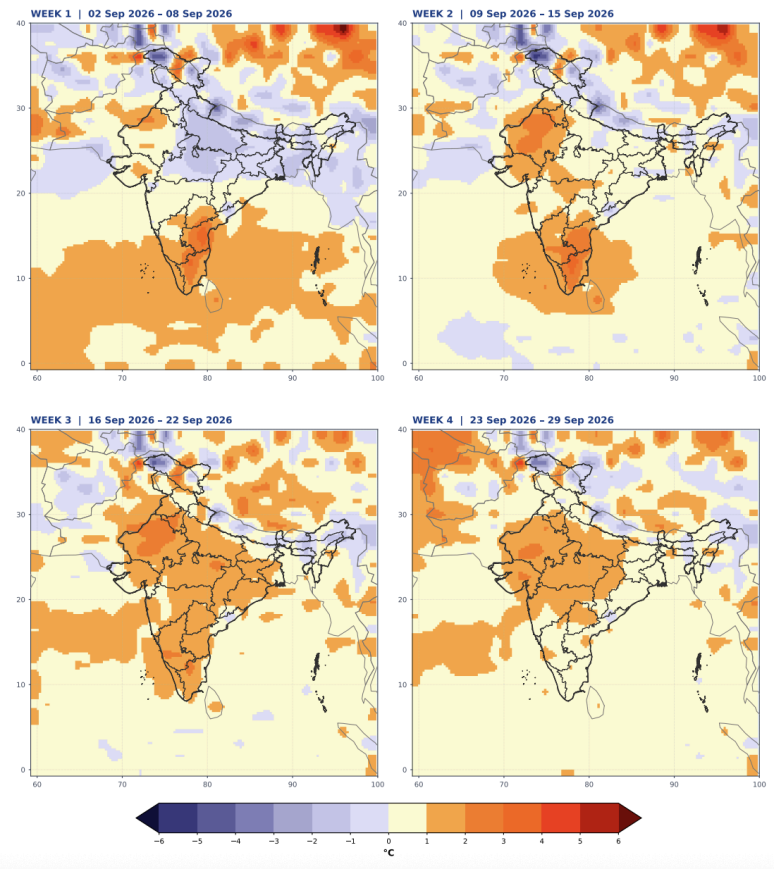



**Figure 3.** Weekly mean temperature anomaly over India for Weeks 1–4, IFS operational initialization dated 2 September 2026. Panels represent (a) 2–8 September, (b) 9–15 September, (c) 16–22 September, and (d) 23–29 September 2026. Anomalies represent departures from the model's typical temperature for the corresponding season and forecast lead during 2002–2021.

**4.1 Week 1: 2–8 September** 2026 - Below-normal temperatures extend across a broad portion of north-central and eastern India, including much of the region experiencing enhanced rainfall. The cool anomaly is physically consistent with enhanced rainfall, increased cloudiness and reduced daytime surface heating. In contrast, southern peninsular India is warmer than normal, with relatively pronounced positive anomalies across portions of Karnataka, Tamil Nadu and adjoining areas. This rainfall-temperature relationship is internally consistent: the anomalously wet north-central corridor is relatively cool, whereas the anomalously dry southern peninsula is relatively warm.

**4.2 Week 2: 9–15 September 2026 -** During Week 2, warmer-than-normal conditions expand substantially. Positive temperature anomalies become pronounced across northwestern and

west-central India and remain strong across the southern peninsula. Conditions over much of central and eastern India become closer to normal, while cooler-than-normal temperatures remain concentrated principally around the Himalayan foothills and portions of Northeast India. This transition accompanies the sharp reduction in rainfall over continental India.

**4.3 Week 3: 16–22 September 2026 -** Week 3 displays the most spatially extensive warm anomaly. Warmer-than-normal temperatures extend across much of western, central, northern and peninsular India, broadly coinciding with widespread below-normal rainfall and substantially weakened low-level monsoon flow. Relatively cooler conditions remain concentrated around portions of the Himalayan region and Northeast India. The Week 3 combination of widespread dry anomalies, positive temperature anomalies and anomalous easterly low-level flow represents the clearest late-monsoon/withdrawal-like signal in this initialization.

**4.4 Week 4: 23–29 September 2026 -** Warm anomalies remain prominent across northwestern, western and north-central India. In contrast, southern and southeastern India become closer to climatological temperatures than during Week 3, consistent with the simultaneous return of modest positive rainfall anomalies in this region. Cooler conditions remain localized near parts of the Himalayas and Northeast India.

**5. Key S2S Signals**

1. **Strong Week 1 north-central/eastern wet anomaly:** The 2–8 September forecast contains a coherent above-normal rainfall corridor from north-central India into eastern India and parts of the Himalayan foothills.
2. **Sharp Week 2 transition:** During 9–15 September, the organized continental wet signal largely disappears and below-normal rainfall becomes widespread across much of India.
3. **Maximum withdrawal-like signal during Week 3:** The combination of negative rainfall anomalies, widespread warm temperature anomalies and weakened/anomalous easterly 850-hPa flow is most clearly expressed during 16–22 September.
4. **Persistent warmth following rainfall suppression:** Southern and western India are warm during Weeks 1–2, with warmer-than-normal conditions expanding across most of central and peninsular India during Week 3.
5. **Northeast India remains comparatively favored:** Northeast India retains intermittent positive rainfall anomalies even as rainfall becomes suppressed across much of the remainder of the country.
6. **Week 4 southern reorganization:** Positive rainfall anomalies return over parts of southern peninsular India and adjoining oceanic regions, accompanied by stronger anomalous low-level flow south of the peninsula.
7. **Withdrawal-like, not a formal withdrawal forecast:** The circulation and rainfall anomalies during Weeks 2–3 are consistent with weakening of the southwest-monsoon circulation, but the model does not establish the official timing or geographical progression of monsoon withdrawal.

## 6. Interpretation

The 2 September 2026 IFS-initialized guidance depicts a two-stage evolution of the Indian monsoon during September. The first stage occurs during 2–8 September, when an active low-level circulation supports a broad zone of enhanced rainfall across north-central and eastern India. The associated region is relatively cool, while the drier southern peninsula is warmer than normal. Week 1 therefore remains dynamically consistent with an active monsoon circulation, although rainfall is concentrated farther north than a spatially uniform active-monsoon pattern. The second stage begins during 9–15 September, when the organized continental wet anomaly rapidly weakens. Low-level westerly anomalies diminish and anomalous easterly components appear over the Arabian Sea and peninsular India. Rainfall becomes below normal over much of western, central and southern India. These characteristics become even more pronounced during 16–22 September. Week 3 contains widespread continental rainfall suppression, anomalously warm temperatures and a broad reduction in the strength of the climatological monsoon circulation. Among the four forecast weeks, Week 3 provides the clearest indication of a transition toward withdrawal-like conditions. The Week 4 forecast introduces additional complexity. During 23–29 September, rainfall remains suppressed over portions of northern India but becomes above normal across parts of southern peninsular India and adjoining oceanic regions. Enhanced anomalous low-level flow also develops south of the peninsula. The Week 4 pattern therefore suggests redistribution and possible episodic reactivation of rainfall over southern India rather than a simple, uninterrupted nationwide retreat of monsoon activity. Accordingly, the 2 September initialization supports a progressive weakening of the core southwest-monsoon circulation after Week 1, strongest during Weeks 2–3, but does not support assigning a formal withdrawal date. Confidence is highest for Week 1, moderate for Week 2 and progressively lower during Weeks 3–4.

## 7. Important Update Relative to the 26 August Initialization

One particularly important forecast change emerges when the new initialization is compared with the 26 August guidance. For the overlapping 2–8 September period, the earlier 26 August forecast treated this as Week 2 and indicated that positive rainfall anomalies would largely become confined to the Himalayan foothills and Northeast India, with much of the rest of India below normal. The 2 September initialization substantially revises that signal. Its Week 1 forecast now shows a coherent positive rainfall anomaly extending across north-central and eastern India, together with stronger-than-normal low-level flow over the central monsoon corridor. Therefore, I would not retain the earlier statement that most of India will be under negative rainfall anomalies during 2–8 September. The updated initialization instead favors an active rainfall episode across a north-central/eastern belt. After this revised Week 1, however, the broader late-September evolution remains broadly consistent with the previous cycle: suppressed rainfall over much of western/central India, warming, and weakening of low-level monsoon flow during the middle of September, followed by a more uncertain and spatially mixed signal later in the month. That forecast-cycle change is scientifically important and worth

highlighting in the published report because it demonstrates why the S2S guidance should be updated with each initialization rather than interpreted as a static four-week prediction.